1

Introduction to Probability

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Notation

$\mathcal{M}_+(X, \mathcal{A})$	finite measures on $(X, \mathcal{A})$
v	velocity

Preface

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Chapter 1

Probability

1.1 Distributions

Definition 1.1.0.1. Let Ω be a set and $\mathcal{P} \subset \mathcal{P}(X)$. Then $\mathcal{P}$ is said to be a **π-system** on Ω if for each $A, B \in \mathcal{P}$, $A \cap B \in \mathcal{P}$.

Definition 1.1.0.2. Let Ω be a set and $\mathcal{L} \subset \mathcal{P}(\Omega)$. Then $\mathcal{L}$ is said to be a **λ-system** on Ω if

1. $\mathcal{L} \neq \varnothing$
2. for each $A \in \mathcal{L}$, $A^c \in \mathcal{L}$
3. for each $(A_n)_{n\in\mathbb{N}} \subset \mathcal{L}$, if $(A_n)_{n\in\mathbb{N}}$ is disjoint, then $\bigcup\limits_{n\in\mathbb{N}} A_n \in \mathcal{L}$

Exercise 1.1.0.3. Let Ω be a set and $\mathcal{L}$ a λ-system on Ω. Then

1. $\Omega, \varnothing \in \mathcal{L}$

Proof. Straightforward. □

Definition 1.1.0.4. Let Ω be a set and $\mathcal{C} \subset \mathcal{P}(\Omega)$. Put

$$\mathcal{S} = \{\mathcal{L} \subset \mathcal{P}(\Omega) : \mathcal{L} \text{ is a } \lambda\text{-system on } \Omega \text{ and } \mathcal{C} \subset \mathcal{L}\}$$

We define the **λ-system on Ω generated by** $\mathcal{C}$, $\lambda(\mathcal{C})$, to be

$$\lambda(\mathcal{C}) = \bigcap_{\mathcal{L}\in\mathcal{S}} \mathcal{L}$$

Exercise 1.1.0.5. Let Ω be a set and $\mathcal{C} \subset \mathcal{P}(\Omega)$. If $\mathcal{C}$ is a λ-system and $\mathcal{C}$ is a π-system, then $\mathcal{C}$ is a σ-algebra.

Proof. Suppose that $\mathcal{C}$ is a λ-system and $\mathcal{C}$ is a π-system. Then we need only verify the third axiom in the definition of a σ-algebra. Let $(A_n)_{n\in\mathbb{N}} \subset \mathcal{C}$. Define $B_1 = A_1$ and for $n \geq 2$, define $B_n = A_n \cap \left(\bigcup\limits_{k=1}^{n-1} A_k\right)^c = A_n \cap \left(\bigcap\limits_{k=1}^{n-1} A_k^c\right) \in \mathcal{C}$. Then $(B_n)_{n\in\mathbb{N}}$ is disjoint and therefore $\bigcup\limits_{n\in\mathbb{N}} A_n = \bigcup\limits_{n\in\mathbb{N}} B_n \in \mathcal{C}$. □

Theorem 1.1.0.6. (Dynkin's Theorem)
Let Ω be a set.

1. Let $\mathcal{P}$ be a π-system on Ω and $\mathcal{L}$ a λ-system on Ω. If $\mathcal{P} \subset \mathcal{L}$, then $\sigma(\mathcal{P}) \subset \mathcal{L}$.
2. Let $\mathcal{P}$ be a π-system on Ω. Then $\sigma(\mathcal{P}) = \lambda(\mathcal{P})$

Exercise 1.1.0.7. Let $(\Omega, \mathcal{F})$ be a measurable space and μ, ν probability measures on $(\Omega, \mathcal{F})$. Put $\mathcal{L}_{\mu,\nu} = \{A \in \mathcal{F} : \mu(A) = \nu(A)\}$. Then $\mathcal{L}_{\mu,\nu}$ is a λ-system on Ω.

Proof.

1. $\varnothing \in \mathcal{L}_{\mu,\nu}$.

2. Let $A \in \mathcal{L}_{\mu,\nu}$. Then $\mu(A) = \nu(A)$. Thus

$$\begin{aligned}\mu(A^c) &= 1 - \mu(A) \\ &= 1 - \nu(A) \\ &= \nu(A^c)\end{aligned}$$

   So $A^c \in \mathcal{L}_{\mu,\nu}$.

3. Let $(A_n)_{n\in\mathbb{N}} \subset \mathcal{L}_{\mu,\nu}$. So for each $n \in \mathbb{N}$, $\mu(A_n) = \nu(A_n)$. Suppose that $(A_n)_{n\in\mathbb{N}}$ is disjoint. Then

$$\begin{aligned}\mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) &= \sum_{n\in\mathbb{N}} \mu(A_n) \\ &= \sum_{n\in\mathbb{N}} \nu(A_n) \\ &= \nu\left(\bigcup_{n\in\mathbb{N}} A_n\right)\end{aligned}$$

   Hence $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{L}_{\mu,\nu}$.

□

Exercise 1.1.0.8. Let $(\Omega, \mathcal{F})$ be a measurable space, μ, ν probability measures on $(\Omega, \mathcal{F})$ and $\mathcal{P} \subset \mathcal{F}$ a π-system on Ω. Suppose that for each $A \in \mathcal{P}$, $\mu(A) = \nu(A)$. Then for each $A \in \sigma(\mathcal{P})$, $\mu(A) = \nu(A)$.

Proof. Using the previous exercise, we see that $\mathcal{P} \subset \mathcal{L}_{\mu,\nu}$. Dynkin's theorem implies that $\sigma(\mathcal{P}) \subset \mathcal{L}_{\mu,\nu}$. So for each $A \in \sigma(\mathcal{P})$, $\mu(A) = \nu(A)$. □

Definition 1.1.0.9. Let $F : \mathbb{R} \to \mathbb{R}$. Then F is said to be a **probability distribution function** if

1. F is right continuous

2. F is increasing

3. $F(-\infty) = 0$ and $F(\infty) = 1$

Definition 1.1.0.10. Let P be a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. We define $F_P : \mathbb{R} \to \mathbb{R}$, by

$$F_P(x) = P((-\infty, x])$$

We call F_P the **probability distribution function of** P.

Exercise 1.1.0.11. Let $(\Omega, \mathcal{F}, P)$ be a probability measure. Then F_P is a probability distribution function.

Proof.

1. Let $x \in \mathbb{R}$ and $(x_n)_{n\in\mathbb{N}} \subset [x, \infty)$. Suppose that $x_n \to x$. Then $(x, x_n] \to \varnothing$ because $\limsup_{n\to\infty}(x, x_n] = \varnothing$. Thus

$$F(x_n) - F(x) = P((x, x_n]) \to P(\varnothing) = 0$$

   This implies that

$$F(x_n) \to F(x)$$

   So F is right continuous.

2. Clearly F_P is increasing.

3. Continuity from below tells us that

$$F(-\infty) = \lim_{n\to-\infty} F(n) = \lim_{n\to-\infty} P((-\infty, n]) = 0$$

and continuity from above tell us that

$$F(\infty) = \lim_{n\to\infty} F(n) = \lim_{n\to\infty} P((-\infty, n]) = 1$$

□

Exercise 1.1.0.12. Let μ, ν be probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Then $F_\mu = F_\nu$ iff $\mu = \nu$.

Proof. Clearly if $\mu = \nu$, then $F_\mu = F_\nu$. Conversely, suppose that $F_\mu = F_\nu$. Then for each $x \in \mathbb{R}$,

$$\begin{aligned}\mu((-\infty, x]) &= F_\mu(x)\\ &= F_\nu(x)\\ &= \nu((-\infty, x])\end{aligned}$$

Put $\mathcal{C} = \{(-\infty, x] : x \in \mathbb{R}\}$. Then $\mathcal{C}$ is a π-system and for each $A \in \mathcal{C}$, $\mu(A) = \nu(A)$. Hence for each $A \in \sigma(C) = \mathcal{B}(\mathbb{R})$, $\mu(A) = \nu(A)$. So $\mu = \nu$. □

Definition 1.1.0.13. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X : \Omega \to \mathbb{R}^n$. Then X is said to be a **random vector** on $(\Omega, \mathcal{F})$ if X is $\mathcal{F}$-$\mathcal{B}(\mathbb{R}^n)$ measurable. If $n = 1$, then X is said to be a **random variable**. We define

$$L^0_n(\Omega, \mathcal{F}, P) = \{X : \Omega \to \mathbb{R}^n : X \text{ is a random vector}\}$$

and

$$L^p_n(\Omega, \mathcal{F}, P) = \left\{X \in L^0_n : \int \|X\|^p dP < \infty\right\}$$

Definition 1.1.0.14. Let $(\Omega, \mathcal{F}, P)$ be a probability space and X a random variable on $(\Omega, \mathcal{F})$. We define the **probability distribution** of X, $P_X : \mathcal{B}(R) \to [0, 1]$, to be the measure

$$P_X = X_* P$$

That is, for each $A \in \mathcal{B}(\mathbb{R})$,

$$P_X(A) = P(X^{-1}(F))$$

We define the **probability distribution function** of X, $F_X : \mathbb{R} \to [0, 1]$, to be

$$F_X = F_{P_X}$$

Definition 1.1.0.15. Let $(\Omega, \mathcal{F}, P)$ be a probability space and X a random variable on $(\Omega, \mathcal{F})$. If $P_X \ll m$, we define the **probability density** of X, $f_X : \mathbb{R} \to \mathbb{R}$, by

$$f_X = \frac{dP_X}{dm}$$

Exercise 1.1.0.16. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables on $(\Omega, \mathcal{F})$. Then for each $x \in \mathbb{R}$,

$$¶\left(\liminf_{n\to\infty} X_n > x\right) \le \liminf_{n\to\infty} P(X_n > x)$$

Proof. Let $\omega \in \left\{\liminf_{n\to\infty} X_n > x\right\}$. Then $x < \liminf_{n\to\infty} X_n(\omega) = \sup_{n\in\mathbb{N}}\left(\inf_{k\ge n} X_k(\omega)\right)$. So there exists $n^* \in \mathbb{N}$ such that $x < \inf_{k\ge n^*} X_k(\omega)$. Then for each $k \in \mathbb{N}$, $k \ge n^*$ implies that $x < X_k(\omega)$. So there exists $n^* \in \mathbb{N}$ such that for each $k \in \mathbb{N}$, $k \ge n^*$ implies that $\mathbf{1}_{\{X_k>x\}}(\omega) = 1$. Hence $\inf_{k\ge n^*} \mathbf{1}_{\{X_k>x\}}(\omega) = 1$. Thus $\liminf_{n\to\infty} \mathbf{1}_{\{X_k>x\}}(\omega) = \sup_{n\in\mathbb{N}}\left(\inf_{k\ge n} \mathbf{1}_{\{X_k>x\}}(\omega)\right) = 1$. Therefore $\omega \in \liminf_{n\to\infty}\{X_k > x\}$ and we have shown that

$$\left\{\liminf_{n\to\infty} X_n > x\right\} \subset \liminf_{n\to\infty}\{X_k > x\}$$

Then

$$\begin{aligned} P\left(\liminf_{n\to\infty} X_n > x\right) &\leq P\left(\liminf_{n\to\infty}\{X_k > x\}\right) \\ &\leq \liminf_{n\to\infty} P(\{X_k > x\}) \end{aligned}$$

□

Definition 1.1.0.17. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X \in L^+(\Omega) \cup L^1$. Define the **expectation of X**, $E(X)$, to be

$$E(X) = \int X \, dP$$

.

1.2 Independence

Definition 1.2.0.1. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{C} \subset \mathcal{F}$. Then $\mathcal{C}$ is said to be **independent** if for each $(A_i)_{i=1}^n \subset \mathcal{C}$,

$$P\left(\bigcap_{k=1}^n A_k\right) = \prod_{k=1}^n P(A_k)$$

Definition 1.2.0.2. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{C}_1, \cdots, \mathcal{C}_n \subset \mathcal{F}$. Then $\mathcal{C}_1, \cdots, \mathcal{C}_n$ are said to be **independent** if for each $A_1 \in \mathcal{C}_1, \cdots, A_n \in \mathcal{C}_n$, $A_1, \cdots, A_n$ are independent.

Note 1.2.0.3. We will explicitely say that for each $i = 1, \cdots, n$, $\mathcal{C}_i$ is independent when talking about the independence of the elements of $\mathcal{C}_i$ to avoid ambiguity.

Definition 1.2.0.4. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X_1, \cdots, X_2$ random variables on $(\Omega, \mathcal{F})$. Then $X_1, \cdots, X_n$ are said to be **independent** if for each $B_1, \cdots, B_n \in \mathcal{B}(\mathbb{R})$, $X_1^{-1}B_1, \cdots, X_n^{-1}B_n$ are independent.

Exercise 1.2.0.5. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X_1, \cdots, X_n$ random variables on $(\Omega, \mathcal{F})$. Then $X_1, \cdots, X_n$ are independent iff $\sigma(X_1), \cdots, \sigma(X_n)$ are independent.

Proof. Suppose that $X_1, \cdots, X_n$ are independent. Let $A_1, \in \sigma(X_1), \cdots, A_n \in \sigma(A_n)$. Then for each $i = 1, \cdots, n$, there exists $B_i \in \mathcal{B}(\mathbb{R})$ such that $A_i = X_i^{-1}(B_i)$. Then $A_1, \cdots, A_n$ are independent. Hence $\sigma(X_1), \cdots, \sigma(X_n)$ are independent. Conversely, suppose that $\sigma(X_1), \cdots, \sigma(X_n)$ are independent. Let $B_1, \cdots, B_n \in \mathcal{B}(\mathbb{R})$. Then for each $i = 1, \cdots, n$, $X_i^{-1}B_i \in \sigma(X_i)$. Then $X_1^{-1}B_1, \cdots, X_n^{-1}B_n$ are independent. Hence $X_1, \cdots, X_n$ are independent. □

Exercise 1.2.0.6. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X_1, \cdots, X_n$ random variables on $(\Omega, \mathcal{F})$ and $\mathcal{F}_1, \cdots, \mathcal{F}_n \subset \mathcal{F}$ a collection of σ-algebras on Ω. Suppose that for each $i = 1, \cdots, n$, X_i is $\mathcal{F}_i$-measurable. If $\mathcal{F}_1, \cdots, \mathcal{F}_n$ are independent, then $X_1, \cdots, X_n$ are independent.

Proof. For each $i = 1, \cdots, n$, $\sigma(X_i) \subset \mathcal{F}_i$. So $\sigma(X_1), \cdots, \sigma(X_n)$ are independent. Hence $X_1, \cdots, X_n$ are independent. □

Exercise 1.2.0.7. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{C}_1, \cdots, \mathcal{C}_n \subset \mathcal{F}$. Suppose that for each $i = 1, \cdots, n$, $\mathcal{C}_i$ is a π-system and $\mathcal{C}_1, \cdots, \mathcal{C}_n$ are independent, then $\sigma(\mathcal{C}_1), \cdots, \sigma(\mathcal{C}_n)$ are independent.

Proof. Let $A_2 \in \mathcal{C}_2$. Define $\mathcal{L} = \{A \in \mathcal{F} : P(A \cap A_2) = P(A)P(A_2)\}$. Then

1. $\Omega \in \mathcal{L}$
2. If $A \in \mathcal{L}$, then

$$\begin{aligned} P(A^c \cap A_2) &= P(A_2) - P(A_2 \cap A) \\ &= P(A_2) - P(A_2)P(A) \\ &= (1 - P(A))P(A_2) \\ &= P(A^c)P(A_2) \end{aligned}$$

So $A^c \in \mathcal{L}$

3. If $(B_n)_{n\in\mathbb{N}} \subset \mathcal{L}$ is disjoint, then

$$\begin{aligned} P\left(\left[\bigcup_{n\in\mathbb{N}} B_n\right] \cap A_2\right) &= P\left(\bigcup_{n\in\mathbb{N}} B_n \cap A_2\right) \\ &= \sum_{n\in\mathbb{N}} P(B_n \cap A_2) \\ &= \sum_{n\in\mathbb{N}} P(B_n)P(A_2) \\ &= \left[\sum_{n\in\mathbb{N}} P(B_n)\right] P(A_2) \\ &= P\left(\bigcup_{n\in\mathbb{N}} A_n\right) P(A_2) \end{aligned}$$

So $\bigcup_{n\in\mathbb{N}} B_n \in \mathcal{L}$.

Thus $\mathcal{L}$ is a λ-system. Since $\mathcal{C}_1 \subset \mathcal{L}$ is a π-system, Dynkin's theorem tells us that $\sigma(\mathcal{C}_1) \subset \mathcal{L}$. Since $A_2 \in \mathcal{C}_2$ is arbitrary $\sigma(\mathcal{C}_1)$ and $\mathcal{C}_2$ are independent. The same reasoning implies that $\sigma(\mathcal{C}_1)$ and $\sigma(\mathcal{C}_2)$ are independent. Let $A_2 \in \mathcal{C}_1, \cdots, A_n \in \mathcal{C}_n$ We may do the same process with

$$\mathcal{L} = \left\{A \in \mathcal{F} : P\left(A \cap \left(\bigcap_{i=2}^{n} A_i\right)\right) = P(A)\prod_{i=2}^{n} P(A_i)\right\}$$

and conclude that $\sigma(\mathcal{C}_1), \mathcal{C}_2, \cdots, \mathcal{C}_n$ are independent. Which, using the same reasoning would imply that $\sigma(\mathcal{C}_1), \cdots, \sigma(\mathcal{C}_n)$ are independent. □

Exercise 1.2.0.8. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X_1, \cdots, X_n$ random variables on $(\Omega, \mathcal{F})$. Then $X_1, \cdots, X_n$ are independent iff for each $x_1, \cdots, x_n \in \mathbb{R}$,

$$P(X_1 \leq x_1, \cdots, X_n \leq x_n) = \prod_{i=1}^{n} P(X_i \leq x_i)$$

Proof. Suppose that $X_1, \cdots, X_n$ are independent. Then $\sigma(X_1), \cdots, \sigma(X_n)$ are independent. Let $x_1, \cdots, x_n \in \mathbb{R}$. Then for each $i = 1, \cdots, n$, $\{X_i \leq x_i\} \in \sigma(X_i)$. Hence
$P(X_1 \leq x_1, \cdots, X_n \leq x_n) = \prod_{i=1}^{n} P(X_i \leq x_i)$. Conversely, suppose that for each
$x_1, \cdots, x_n \in \mathbb{R}$, $P(X_1 \leq x_1, \cdots, X_n \leq x_n) = \prod_{i=1}^{n} P(X_i \leq x_i)$. Define $\mathcal{C} = \{(-\infty, x] : x \in \mathbb{R}\}$. Then $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C})$. For each $i = 1, \cdots, n$, define $\mathcal{C}_i = X_i^{-1}\mathcal{C}$. Then for each $i = 1, \cdots, n$, $\mathcal{C}_i$ is a π-system and

$$\begin{aligned} \sigma(\mathcal{C}_i) &= \sigma(X^{-1}(\mathcal{C})) \\ &= X_i^{-1}(\sigma(\mathcal{C})) \\ &= X_i^{-1}(\mathcal{B}(\mathbb{R})) \\ &= \sigma(X_i) \end{aligned}$$

By assumption, $\mathcal{C}_1, \cdots, \mathcal{C}_n$ are independent. The previous exercies tells us that $\sigma(X_1), \cdots, \sigma(X_n)$ are independent. Then $X_1, \cdots, X_n$ are independent. □

Exercise 1.2.0.9. Let Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X_1, \cdots, X_n$ random variables on $(\Omega, \mathcal{F})$. Define $X = (X_1, \cdots, X_n)$. If $X_1, \cdots, X_n$ are independent, then

$$P_X = \prod_{i=1}^{n} P_{X_i}$$

.

Proof. Let $A_1, \cdots, A_n \in \mathcal{B}(\mathbb{R})$. Then

$$\begin{aligned} P_X(A_1 \times \cdots \times A_n) &= P(X \in A_1 \times \cdots \times \in A_n) \\ &= P(X_1 \in A_1, \cdots, X_n \in A_n) \\ &= P(X_1 \in A_1) \cdots P(X_n \in A_n) \\ &= P_{X_1}(A_1) \cdots P_{X_n}(A_n) \\ &= \prod_{i=1}^{n} P_{X_i}(A_1 \times \cdots \times A_n) \end{aligned}$$

Put

$$\mathcal{P} = \{A_1 \times \cdots \times A_n : A_1 \in \mathcal{B}(R), \cdots, A_n \in \mathcal{B}(R)\}$$

Then $\mathcal{P}$ is a π-system and

$$\sigma(\mathcal{P}) = \mathcal{B}(R) \otimes \cdots \otimes \mathcal{B}(R) = \mathcal{B}(\mathbb{R}^n)$$

A previous exercise then tells us that $P_X = \prod_{i=1}^{n} P_{X_i}$ □

Exercise 1.2.0.10. Let Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X_1, \cdots, X_n$ random variables on $(\Omega, \mathcal{F})$ and $f_1, \cdots, f_n : \mathbb{R} \to \mathbb{R} \in L^0$. Suppose that $f_1 \circ X_1, \cdots, f_n \circ X_n \in L^+(\Omega)$ or $f_1 \circ X_1, \cdots, f_n \circ X_n \in L^1(\Omega)$. If $X_1, \cdots, X_n$ are independent, then

$$E(f_1(X_1) \cdots f_n(X_n)) = \prod_{i=1}^{n} E(f_i(X_i))$$

Proof. Define the random vector $X : \Omega \to \mathbb{R}^n$ by $X = (X_1, \cdots, X_n)$ and $g : \mathbb{R}^n \to \mathbb{R}$ by $g(x_1, \cdots, x_n) = f_1(x_1) \cdots f_n(x_n)$. Suppose that for each $i = 1, \cdots, n$, $f_i \in L^+(\mathbb{R})$. Then $g \in L^+(\mathbb{R}^n)$ and by change of variables,

$$\begin{aligned} E(f_1(X_1) \cdots f_n(X_n)) &= E(g(X)) \\ &= \int_\Omega g \circ X \, dP \\ &= \int_{\mathbb{R}^n} g(x) \, dP_X(x) \\ &= \int_{R^n} g(x) d \prod_{i=1}^{n} P_{X_i}(x) \\ &= \prod_{i=1}^{n} \int_{\mathbb{R}} f_i(x) dP_{X_i}(x) \\ &= \prod_{i=1}^{n} \int_\Omega f_i \circ X dP \\ &= \prod_{i=1}^{n} E(f_i(X_i)) \end{aligned}$$

If for each $i = 1, \cdots, n$, $f_i \in L^1(\mathbb{R}, P_{X_i})$, then following the above reasoning with $|g|$ tells us that $g \in L^1(\mathbb{R}^n, P_X)$ and we use change of variables and Fubini's theorem to get the same result. □

1.3 L^p Spaces for Probability

Note 1.3.0.1. Recall that for a probability space $(\Omega, \mathcal{F}, P)$ and $1 \leq p \leq q \leq \infty$ we have $L^q \subset L^p$ and for each $X \in L^q$, $\|X\|_p \leq \|X\|_q$. Also recall that for $X, Y \in Ł^2$, we have that $\|XY\|_1 \leq \|X\|_2 \|X\|_2$.

Definition 1.3.0.2. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X \in L^2$. Define the **variance of X**, $Var(X)$, to be

$$Var(X) = E(\big[\big(X - E(X)\big)^2\big)$$

.

Definition 1.3.0.3. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X, Y \in L^2$. Define the

Definition 1.3.0.4. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X, Y \in L^2$. Define the **covariance of** X **and** Y, $Cov(X, Y)$, to be

$$Cov(X, Y) = E([X - E(X)][Y - E(Y)])$$

Exercise 1.3.0.5. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X, Y \in L^2$. Then the covariance is well defined and $Cov(X, Y)^2 \leq Var(X)Var(Y)$

Proof. By Holder's inequality,

$$\begin{aligned}
|Cov(X, Y)| &= \left| \int (X - E(X))(Y - E(Y))\, dP \right| \\
&\leq \int |(X - E(X))(Y - E(Y))|\, dP \\
&= \|(X - E(X))(Y - E(Y))\|_1 \\
&\leq \|X - E(X)\|_2 \|(Y - E(Y)\|_2 \\
&= \left(\int |X - E(X)|^2\, dP \right)^{\frac{1}{2}} \left(|Y - E(Y)|^2 \right)^{\frac{1}{2}} \\
&= Var(X)^{\frac{1}{2}} Var(Y)^{\frac{1}{2}}
\end{aligned}$$

So $Cov(X, Y)^2 \leq Var(X)Var(Y)$. □

Exercise 1.3.0.6. Let $(\Omega, \mathcal{F}, P)$ be a measure space and $X, Y \in L^2$. Then

1. $Cov(X, Y) = E(XY) - E(X)E(Y)$
2. If X, Y are independent, then $Cov(X, Y) = 0$
3. $Var(X) = E(X^2) - E(X)^2$
4. for each $a, b \in \mathbb{R}$, $Var(aX + b) = a^2 Var(X)$.
5. $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$

Proof.

1. We have that

$$\begin{aligned}
Cov(X, Y) &= E\Big[(X - E(X))(Y - E(Y))\Big] \\
&= E(XY - E(Y)X - E(X)Y + E(X)E(Y)) \\
&= E(XY) - E(X)E(Y) - E(X)E(Y) + E(X)E(Y) \\
&= E(XY) - E(X)E(Y)
\end{aligned}$$

2. Suppose that X, Y are independent. Then $E(XY) = E(X)E(Y)$. Hence

$$\begin{aligned}
Cov(X, Y) &= E(XY) - E(X)E(Y) \\
&= E(X)E(Y) - E(X)E(Y) \\
&= 0
\end{aligned}$$

3. Part (1) implies that

$$\begin{aligned}
Var(X) &= Cov(X, X) \\
&= E(X^2) - E(X)^2
\end{aligned}$$

4. Let $a, b \in \mathbb{R}$. Then

$$\begin{aligned}
Var(aX+b) &= E[(aX+b)^2] - E(aX+b)^2 \\
&= E[a^2X^2 + 2abX + b^2] - (aE(X)+b)^2 \\
&= a^2E(X^2) + 2abE(X) + b^2 - (a^2E(X)^2 + 2abE(X) + b^2) \\
&= a^2(E(X^2) - E(X)^2) \\
&= a^2Var(X)
\end{aligned}$$

5. We have that

$$\begin{aligned}
Var(X+Y) &= E[(X+Y)^2] - E[X+Y]^2 \\
&= E[X^2 + 2XY + Y^2] - (E(X) + E[Y])^2 \\
&= E(X^2) + 2E[XY] + E[Y^2] - (E(X)^2 + 2E(X)E[Y] + E[Y]^2) \\
&= (E(X^2) - E(X)^2) + (E[Y^2] - E[Y]^2) + 2(E[XY] - E(X)E[Y]) \\
&= Var(X) + Var(Y) + 2Cov(X,Y)
\end{aligned}$$

□

Definition 1.3.0.7. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X, Y \in L^2$. The **correlation of X and Y**, $Cor(X,Y)$, is defined to be

$$Cor(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}}$$

Exercise 1.3.0.8.

Exercise 1.3.0.9. Jensen's Inequality:
Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X \in L^1$ and $\phi : \mathbb{R} \to \mathbb{R}$. If ϕ is convex, then

$$\phi(E(X)) \leq E[\phi(X)]$$

Proof. Put $x_0 = E(X)$. Since ϕ is convex, there exist $a, b \in \mathbb{R}$ such that $\phi(x_0) = ax_0 + b$ and for each $x \in \mathbb{R}$, $\phi(x) \geq ax + b$. Then

$$\begin{aligned}
E[\phi(X)] &= \int \phi(X)\, dP \\
&\geq \int [aX + b]\, dP \\
&= a\int X\, dP + b \\
&= aE(X) + b \\
&= ax_0 + b \\
&= \phi(x_0) \\
&= \phi(E(X))
\end{aligned}$$

□

Exercise 1.3.0.10. Markov's Inequality: Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X \in L^+$. Then for each $a \in (0, \infty)$,

$$P(X \geq a) \leq \frac{E(X)}{a}$$

Proof. Let $a \in (0, \infty)$. Then $a\mathbf{1}_{\{X \geq a\}} \leq X\mathbf{1}_{\{X \geq a\}}$. Thus

$$\begin{aligned} aP(X \geq a) &= \int a\mathbf{1}_{\{X \geq a\}}\, dP \\ &= \int X\mathbf{1}_{\{X \geq a\}}\, dP \\ &\leq \int X\, dP \\ &= E(X) \end{aligned}$$

Therefore

$$P(X \geq a) \leq \frac{E(X)}{a}$$

. □

Exercise 1.3.0.11. Chebychev's Inequality: Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X \in L^2$. Then for each $a \in (0, \infty)$,

$$P(|X - E(X)| \geq a) \leq \frac{Var(X)}{a^2}$$

Proof. Let $a \in (0, \infty)$. Then

$$\begin{aligned} P(|X - E(X)| \geq a) &= P((X - E(X))^2 \geq a^2) \\ &\leq \frac{E[(X - E(X))^2]}{a^2} \\ &= \frac{Var(X)}{a^2} \end{aligned}$$

□

Exercise 1.3.0.12. Chernoff's Bound: Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X \in L^2$. Then for each $a, t \in (0, \infty)$,

$$P(X \geq a) \leq e^{-ta} E[e^{tX}]$$

Proof. Let $a, t \in (0, \infty)$. Then

$$\begin{aligned} P(X \geq a) &= P(tX \geq ta) \\ &= P(e^{tX} \geq e^{ta}) \\ &\leq e^{-ta} E[e^{tX}] \end{aligned}$$

□

Exercise 1.3.0.13. Weak Law of Large Numbers: Let $(\Omega, \mathcal{F}, P)$ be a probability space $(X_i)_{i \in \mathbb{N}} \subset L^2$. Suppose that $(X_i)_{i \in \mathbb{N}}$ are iid. Then

$$\frac{1}{n} \sum_{i=1}^{n} X_i \xrightarrow{P} E[X_1]$$

Proof. Put $\mu = E[X_1]$ and $\sigma^2 = Var(X_1)$. Then

$$\begin{aligned} E[\frac{1}{n} \sum_{i=1}^{n} X_i] &= \frac{1}{n} \sum_{i=1}^{n} E[X_i] \\ &= \frac{1}{n} \sum_{i=1}^{n} \mu \\ &= \mu \end{aligned}$$

and

$$\begin{aligned}
Var(\frac{1}{n}\sum_{i=1}^{n} X_i) &= \frac{1}{n^2} Var(\sum_{i=1}^{n} X_i) \\
&= \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) \\
&= \frac{1}{n^2}\sum_{i=1}^{n} \sigma^2 \\
&= \frac{\sigma^2}{n}
\end{aligned}$$

Let $\epsilon > 0$. Then

$$\begin{aligned}
P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - E[X_1]\right| \geq \epsilon\right) &= P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right| \geq \epsilon\right) \\
&= P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - E(\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]\right| \geq \epsilon\right) \\
&\leq \frac{Var(\frac{1}{n}\sum_{i=1}^{n} X_i)}{\epsilon^2} \\
&= \frac{\sigma^2/n}{\epsilon^2} \\
&= \frac{\sigma^2}{n\epsilon^2} \to 0
\end{aligned}$$

So

$$\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{P} E[X_1]$$

□

1.4 Borel Cantelli Lemma

Exercise 1.4.0.1. Borel Cantelli Lemma:
Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(A_n)_{n\in\mathbb{N}} \subset \mathcal{F}$.

1. If $\sum_{n\in\mathbb{N}} P(A_n) < \infty$, then $P(\limsup_{n\to\infty} A_n) = 0$.
2. If $(A_n)_{n\in\mathbb{N}}$ are independent and $\sum_{n\in\mathbb{N}} P(A_n) = \infty$, then $P(\limsup_{n\to\infty} A_n) = 1$

Proof.

1. Suppose that $\sum_{n\in\mathbb{N}} P(A_n) < \infty$. Recall that
$$\limsup_{n\to\infty} A_n = \left\{\omega \in \Omega : \sum_{n\in\mathbb{N}} 1_{A_n}(\omega) = \infty\right\}$$
Then
$$\begin{aligned}\infty &> \sum_{n\in\mathbb{N}} P(A_n)\\ &= \sum_{n\in\mathbb{N}} \int 1_{A_n}\, dP\\ &= \int \sum_{n\in\mathbb{N}} 1_{A_n}\, dP\end{aligned}$$
Thus $\sum_{n\in\mathbb{N}} 1_{A_n} < \infty$ a.e. and $P(\limsup_{n\to\infty} A_n) = 0$.
2. Suppose that $(A_n)_{n\in\mathbb{N}}$ are independent and $\sum_{n\in\mathbb{N}} P(A_n) = \infty$.

□

Exercise 1.4.0.2. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(X_n)_{n\in\mathbb{N}} \subset L^0$ and $X \in L^0$.

1. If for each $\epsilon > 0$, $\sum_{n\in\mathbb{N}} P(|X_n - X| \geq \epsilon) < \infty$, then $X_n \to X$ a.e.
2. If $(X_n)_{n\in\mathbb{N}}$ are independent and there exists $\epsilon > 0$ such that $\sum_{n\in\mathbb{N}} P(|X_n - X| \geq \epsilon) = \infty$, then $X_n \not\to X$ a.e.

Proof.

1. For $\epsilon > 0$ and $n \in \mathbb{N}$, set $A_n(\epsilon) = \{\omega \in \Omega : |X_n(\omega) - X(\omega)| \geq \epsilon\}$. Suppose that for each $\epsilon > 0$, $\sum_{n\in\mathbb{N}} P(|X_n - X| \geq \epsilon) < \infty$. The Borel-Cantelli lemma implies that for each $m \in \mathbb{N}$,
$$P(\limsup_{n\to\infty} A_n(1/m)) = 0$$
Let $\omega \in \Omega$. Then $X_n(\omega) \not\to X(\omega)$ iff
$$\omega \in \bigcup_{m\in\mathbb{N}} \limsup_{n\to\infty} A_n(1/m)$$
So
$$\begin{aligned}P(X_n \not\to X) &= P\left(\bigcup_{m\in\mathbb{N}} \limsup_{n\to\infty} A_n(1/m)\right)\\ &\leq \sum_{m\in\mathbb{N}} P(\limsup_{n\to\infty} A_n(1/m))\\ &= 0\end{aligned}$$
Hence $X_n \to X$ a.e.

2.

□

Chapter 2

Probability on locally compact Groups

Note 2.0.0.1. In this section, familiarity with Haar measure will be assumed. This section is intended as a continuation of section 7 of [4].

2.1 Action on Probability Measures

Note 2.1.0.1. We recall some notation from section 7.1 of [4].

- $l_g \in \text{Homeo}(G)$, $l_g(x) = gc$
- $L_g \in \text{Sym}(L_0(G))$, $L_g f = f \circ l_g^{-1}$ We continue from section 7

Note 2.1.0.2. The next exercise generalizes the notion of a scale-family.

Exercise 2.1.0.3. Let $(\Omega, \mathcal{F}, P)$ be a probability space, G a locally compact group, μ a left Haar measure on G, $X \in L_G^0$ and $g \in G$. If $P_X \ll \mu$, then $f_{gX} = L_g f_X$.

Proof. Suppose that $P_X \ll \mu$. Let $A \in \mathcal{B}(G)$. Then

$$\begin{aligned} P_{gX}(A) &= P(gX \in A) \\ &= P(X \in g^{-1}A) \\ &= P_X(g^{-1}A) \\ &= P_X(l_g^{-1}(A)) \\ &= l_{g*}P_X(A) \\ &= g \cdot P_X(A) \end{aligned}$$

The previous exercise tells us that $f_{gX} = L_g f_X$. $\square$

Chapter 3

Gaussian Measures

3.1 Gaussian Measures on $\mathbb{R}$

Definition 3.1.0.1. Let $\mu \in \mathcal{M}_1(\mathbb{R})$. Then μ is said to be **Gaussian** if

3.2 Gaussian Measures on Topological Vector Spaces

Definition 3.2.0.1. Let W be a topological vector space and $\mu \in \mathcal{M}_1(W)$. Then μ is said to be **Gaussian** if for each $\phi \in W^*$, $\phi_*\mu$ is Gaussian.

need exercise in analysis notes showing $\mathbb{R}^{\mathbb{N}}$ is a topological vector space.

Definition 3.2.0.2. For $n \in \mathbb{N}$, define $\mu_n \in \mathcal{M}_1(\mathbb{R})$ by

$$\mu_n(A) := \int_A \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dm(x).$$

An exercise in measure theory notes section on projective limits of measure spaces implies that there exists a unique $\mu \in \mathcal{M}_1(\mathbb{R}^{\mathbb{N}})$ such that

- for each $n \in \mathbb{N}$, $\pi_n^*\mu = \mu_n$ and for each,
- for each $(A_n)_{n\in\mathbb{N}} \in \mathcal{B}(\mathbb{R})^{\mathbb{N}}$,

$$\mu\left(\prod_{n\in\mathbb{N}} A_n\right) = \prod_{n\in\mathbb{N}} \mu_n(A_n).$$

We refer to μ as the **standard Gaussian measure** on $\mathbb{R}^{\mathbb{N}}$.

Exercise 3.2.0.3. μ is Gaussian.

Proof. Let $n \in \mathbb{N}$ and $\lambda \in \mathbb{R}$. Define $\eta_\lambda : \mathbb{R}^{\mathbb{N}} \to \mathbb{R}^{\mathbb{N}}$ by $\eta_\lambda(x) := \lambda x$. Then

$$\begin{aligned}(\lambda\pi_n)_*\mu &= (\eta_\lambda \circ \pi_n)_*\mu \\ &= (\eta_\lambda)_*((\pi_n)_*\mu) \\ &= (\eta_\lambda)_*\mu_n\end{aligned}$$

Define $\nu \in \mathcal{M}_1(\mathbb{R})$ by $\nu := (\eta_\lambda)_*\mu_n$. If $\lambda = 0$, then $\nu = \delta_0$. Suppose that $\lambda \neq 0$. Since $\mu_n \ll m$, (check/give details) we have that $\nu \ll m$. The Radon-Nikodym-Lebesgue theorem implies that there exists $f \in L^1(\mathbb{R}, \mathcal{B}(R), m)$ such that $d\nu = f\, dm$. An exercise in analysis notes (need to rework in the newer general framerwork) implies that for m-a.e. $x \in \mathbb{R}$,

$$\widehat{\nu}(x) = \int_{\mathbb{R}} e^{itx}\, d\nu(t) \qquad =$$

$$\begin{aligned}f(x) &= \lim_{r\to 0} \frac{\nu((x-r, x+r))}{m((x-r, x+r))} \\ &= \lim_{r\to 0} \frac{\mu_n(\eta_\lambda^{-1}(x-r, x+r))}{m((x-r, x+r))} \\ &= \lim_{r\to 0} \frac{\mu_n(\lambda^{-1}(x-r, x+r))}{m((x-r, x+r))} \\ &= \lim_{r\to 0} \frac{\mu_n((\lambda^{-1}x - \lambda^{-1}r, \lambda^{-1}x + \lambda^{-1}r))}{m((x-r, x+r))} \\ &= \lim_{r\to 0} \frac{1}{2r} \int_{\lambda^{-1}x-\lambda^{-1}r}^{\lambda^{-1}x+\lambda^{-1}r} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dm(x) \\ &= \lim_{r\to 0} \frac{1}{2r} \int_u\end{aligned}$$

$f(x)$ Let $\phi \in (\mathbb{R}^{\mathbb{N}})^*$. An exercise in the analysis notes section on topological vector spaces subsection on products implies that $(\mathbb{R}^{\mathbb{N}})^* = \operatorname{span}\{\pi_j : j \in \mathbb{N}\}$.

Then show that the pushforward of a linear combos of projection maps is Gaussian $\square$

3.3 Gaussian Measures on Hilbert Spaces

Note 3.3.0.1. recall definitions of $(Y, (\tau_n)_{n\in\mathbb{N}})$ and $((Y_n)_{n\in\mathbb{N}}, (\tau_{n,k})_{(n\leq k)})$ from measure and integration notes section on projective systems of complex measure spaces, specifically the exercise on existence and uniqueness of product measures. We still need an exercise to show that $(Y, (\tau_n)_{n\in\mathbb{N}})$ is a **Top**-projective limit of $((Y_n)_{n\in\mathbb{N}}, (\tau_{n,k})_{(n\leq k)})$ in the analysis notes.

Exercise 3.3.0.2. Let $a \in \mathbb{C}^{\mathbb{N}}$. Then $l^{2,a} \in \mathcal{B}(\mathbb{R}^{\mathbb{N}})$.

Proof. We $\square$

Chapter 4

Weak Convergence of Measures

Chapter 5

Concentration Inequalities

5.1 Introduction

Exercise 5.1.0.1. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X, Y \in L^0_{\mathbb{R}}(\Omega, \mathcal{F}, P)$. Then for each $s, t \in \mathbb{R}$,

$$P(X + Y \geq s + t) \leq P(X \geq s) + P(Y \geq t)$$

Proof. For $Z \in L^0_{\mathbb{R}}(\Omega, \mathcal{F}, P)$ and $t \in \mathbb{R}$, define $A^t_Z \in \mathcal{F}$ by

$$A^t_Z = \{\omega \in \Omega : Z(\omega) \geq t\}$$

Let $s, t \in \mathbb{R}$. Since $(A^s_X)^c \cap (A^t_Y)^c \subset (A^{s+t}_{X+Y})^c$, we have that $A^{s+t}_{X+Y} \subset A^s_X \cup A^t_Y$. Then

$$\begin{aligned} P(X + Y \geq s + t) &= P(A^{s+t}_{X+Y}) \\ &\leq P(A^s_X) + P(A^t_Y) \\ &= P(X \geq s) + P(Y \geq t) \end{aligned}$$

□

Exercise 5.1.0.2. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X, Y \in L^+(\Omega, \mathcal{F}, P)$. Then for each $s, t \geq 0$,

$$P(XY \geq st) \leq P(X \geq s) + P(Y \geq t)$$

Proof. For $Z \in L^0_{\mathbb{R}}(\Omega, \mathcal{F}, P)$ and $t \in \mathbb{R}$, define define $A^t_Z \in \mathcal{F}$ by

$$A^t_Z = \{\omega \in \Omega : Z(\omega) \geq t\}$$

Let $s, t \in \mathbb{R}$. Since $(A^s_X)^c \cap (A^t_Y)^c \subset (A^{st}_{XY})^c$, we have that $A^{st}_{XY} \subset A^s_X \cup A^t_Y$. Then

$$\begin{aligned} P(XY \geq st) &= P(A^{st}_{XY}) \\ &\leq P(A^s_X) + P(A^t_Y) \\ &= P(X \geq s) + P(Y \geq t) \end{aligned}$$

□

5.2 Sub α-Exponential Random Variables

Definition 5.2.0.1. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X \in L^0(\Omega, \mathcal{F}, P)$ and $\alpha > 0$. Then X is said to be **sub α-exponential** if there exist $M, K > 0$ such that for each $t \geq 0$,

$$P(|X| \geq t) \leq Me^{-Kt^\alpha}$$

Exercise 5.2.0.2. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X \in L^0(\Omega, \mathcal{F}, P)$ and $\alpha > 0$. Then the following are equivalent:

1. X is sub α-exponential
2. there exists $K > 0$ such that for each $p \geq 1$, $\|X\|_p \leq Kp^{1/\alpha}$
3.
4.

Proof.

- (1) $\implies$ (2):
  Choose $C_\alpha > 0$ such that for each $x \geq \alpha^{-1}$, $\Gamma(x) \leq C_\alpha x^x$. Since X is sub α-exponential, there exist $M, K_0 > 0$ such that for each $t \geq 0$, $P(|X| \geq t) \leq Me^{-Kt^\alpha}$. Set $K = \max(M\alpha^{-1}C_\alpha, 1)2K_0^{-1/\alpha}\alpha^{-1/\alpha}$. Let $p \geq 1$. Then $p\alpha^{-1} \geq \alpha^{-1}$

$$\begin{aligned}
\|X\|_p^p &= E(|X|^p) \\
&= \int_0^\infty P(|X|^p \geq t)dt \\
&= \int_0^\infty P(|X| \geq t^{1/p})dt \\
&\leq \int_0^\infty Me^{-K_0 t^{\alpha/p}}dt \\
&= Mp\alpha^{-1}\int_0^\infty u^{p/\alpha-1}e^{-K_0 u}du \\
&= Mp\alpha^{-1}\Gamma(p/\alpha)K_0^{-p/\alpha} \\
&\leq Mp\alpha^{-1}C_\alpha(p\alpha^{-1})^{p/\alpha}K_0^{-p/\alpha}
\end{aligned}$$

  Therefore

$$\begin{aligned}
\|X\|_p &\leq (M\alpha^{-1}C_\alpha)^{1/p}p^{1/p}K_0^{-1/\alpha}\alpha^{-1/\alpha}p^{1/\alpha} \\
&\leq \max(M\alpha^{-1}C_\alpha, 1)2K_0^{-1/\alpha}\alpha^{-1/\alpha}p^{1/\alpha} \\
&= Kp^{1/\alpha}
\end{aligned}$$

- (2) $\implies$ (3):

□

Definition 5.2.0.3. Let $\psi : [0, \infty) \to [0, \infty)$. Then ψ is said to be an **Orlicz function** if

1. ψ is convex
2. ψ is increasing
3. $\psi(x) \to \infty$ as $x \to \infty$
4. $\psi(0) = 0$

Definition 5.2.0.4. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\psi : [0, \infty) \to [0, \infty)$ and Orlicz function. We define the **Orlicz ψ-norm**, denoted $\|\cdot\|_\psi : L^0(\Omega, \mathcal{F}, P) \to [0, \infty]$, by

$$\|X\|_\psi = \inf\{t > 0 : E[\psi(|X|/t)] \leq 1\}$$

We define the **Orlicz ψ-space**, denoted $L^\psi(\Omega, \mathcal{F}, P)$, by

$$L^\psi(\Omega, \mathcal{F}, P) = \{X \in L^1(\Omega, \mathcal{F}, P) : \|X\|_\psi < \infty\}$$

Exercise 5.2.0.5. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\psi : [0, \infty) \to [0, \infty)$ an Orlicz function. Then L^ψ is a vector space and $\|\cdot\|_\psi : L^\psi \to [0, \infty)$ is a norm.
Hint: note that

- for $s, t > 0$,
$$\frac{|X|}{s+t} + \frac{|Y|}{s+t} = \frac{s}{s+t}\frac{|X|}{s} + \frac{t}{s+t}\frac{|Y|}{t}$$
- ψ is star-shaped, i.e. for each $x, t \geq 0$ and $f(tx) \leq tf(x)$.

Proof. For $X \in L^0(\Omega, \mathcal{F}, P)$, define $A_X \in \mathcal{F}$ by
$$A_X = \{t > 0 : E[\psi(|X|/t)] \leq 1\}$$
Let $X, Y \in L^\psi(\Omega, \mathcal{F}, P)$ and $\lambda \in \mathbb{C}$. Since $\|X\|_\psi < \infty$ and $\|Y\|_\psi < \infty$, we have that $A_X \neq \varnothing$ and $A_Y \neq \varnothing$,

1. **subadditivity:** Let $\epsilon > 0$. Then there exists $s \in A_X$ and $t \in A_Y$ such that $s < \inf A_X + \epsilon/2$ and $t < \inf A_Y + \epsilon/2$. Since is convex and increasing, we have that
$$\begin{aligned}\psi\left(\frac{|X+Y|}{s+t}\right) &\leq \psi\left(\frac{|X|}{s+t} + \frac{|Y|}{s+t}\right)\\ &= \psi\left(\frac{s}{s+t}\frac{|X|}{s} + \frac{t}{s+t}\frac{|Y|}{t}\right)\\ &\leq \frac{s}{s+t}\psi\left(\frac{|X|}{s}\right) + \frac{t}{s+t}\psi\left(\frac{|Y|}{t}\right)\end{aligned}$$
Therefore,
$$\begin{aligned}E\left[\psi\left(\frac{|X+Y|}{s+t}\right)\right] &\leq \frac{s}{s+t}E\left[\psi\left(\frac{|X|}{s}\right)\right] + \frac{t}{s+t}E\left[\psi\left(\frac{|Y|}{t}\right)\right]\\ &\leq \frac{s}{s+t} + \frac{t}{s+t}\\ &\leq 1\end{aligned}$$
Hence $s + t \in A_{X+Y}$. Thus $A_{X+Y} \neq \varnothing$. Since $s < \inf A_X + \epsilon/2$ and $t < \inf A_Y + \epsilon/2$, we have that
$$\begin{aligned}\|X+Y\|_\psi &= \inf A_{X+Y}\\ &\leq s+t\\ &< \inf A_X + \inf A_Y + \epsilon\\ &= \|X\|_\psi + \|Y\|_\psi + \epsilon\end{aligned}$$
Since $\epsilon > 0$ is arbitrary, $\|X+Y\|_\psi \leq \|X\|_\psi + \|Y\|_\psi$. So $X + Y \in L^\psi(\Omega, \mathcal{F}, P)$
2. **absolute homogeneity:** Suppose that $\lambda = 0$. Then for each $t > 0$,
$$\begin{aligned}E[\psi(|\lambda X|/t)] &= E[\psi(0)]\\ &= E[0]\\ &= 0\\ &\leq 1\end{aligned}$$
Thus
$$\begin{aligned}\|\lambda X\|_\psi &= 0\\ &= |\lambda|\|X\|_\psi\end{aligned}$$

Suppose that $\lambda \neq 0$. Since for each $t > 0$,

$$\psi\left(\frac{|X|}{t}\right) = \psi\left(\frac{|\lambda X|}{|\lambda| t}\right)$$

we have that for each $t > 0$, $t \in A_X$ iff $|\lambda| t \in A_{\lambda X}$. Therefore $A_{\lambda X} = |\lambda| A_X$ and

$$\begin{aligned} \|\lambda X\|_\psi &= \inf A_{\lambda X} \\ &= |\lambda| \inf A_X \\ &= |\lambda| \|X\|_\psi \end{aligned}$$

So $|\lambda X\| \in L^\psi(\Omega, \mathcal{F}, P)$.

3. **positive definiteness:** Note that since ψ is increasing, for each $t_0 > 0$, $t_0 \in A_X$ implies that $[t_0, \infty) \subset A_X$. Suppose that $\|X\|_\psi = 0$. Then $(0, \infty) \subset A_X$. Jensen's inequality implies that for each $t > 0$,

$$\begin{aligned} \psi\left(\frac{E|X|}{t}\right) &\leq E\left[\psi\left(\frac{|X|}{t}\right)\right] \\ &\leq 1 \end{aligned}$$

For the sake of contradiction, suppose that $E|X| > 0$. Since $\psi(x) \to \infty$ as $x \to \infty$, Choose $x > 0$ such that $\psi(x) > 1$. Set $t = E|X|/x$. Then $t > 0$ and

$$\begin{aligned} 1 &< \psi(x) \\ &= \psi\left(\frac{E|X|}{t}\right) \\ &\leq 1 \end{aligned}$$

which is a contradiction. Hence $E|X| = 0$. Thus $X = 0$ a.s.

□

Exercise 5.2.0.6. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\psi : [0, \infty) \to [0, \infty)$ an Orlicz function. Then $L^\psi(\Omega, \mathcal{F}, P)$ is a Banach space.

Proof. □

Chapter 6

Conditional Expectation and Probability

6.1 Conditional Expectation

Exercise 6.1.0.1. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $\mathcal{G}$ a sub σ-algebra of $\mathcal{F}$ and $X \in L^1(\Omega, \mathcal{F}, P)$. Define $P_{\mathcal{G}} = P|_{\mathcal{G}}$ and $Q : \mathcal{G} \to [0, \infty)$ by $Q(G) = \int_G X dP$. Then $Q \ll P_{\mathcal{G}}$.

Proof. Let $G \in \mathcal{G}$. Suppose that $P_{\mathcal{G}}(G) = 0$. By definition, $P(G) = 0$. So $Q(G) = 0$ and $Q \ll P_{\mathcal{G}}$. □

Definition 6.1.0.2. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $\mathcal{G}$ a sub σ-algebra of $\mathcal{F}$ and $X, Y \in L^1(\Omega, \mathcal{F}, P)$. Then Y is said to be a **conditional expectation of X given $\mathcal{G}$** if

1. Y is $\mathcal{G}$-measurable
2. for each $G \in \mathcal{G}$,
$$\int_G Y \, dP = \int_G X \, dP$$

Since (2) implies that conditional expectations of X given $\mathcal{G}$ are equal $P_{\mathcal{G}}$-a.e., we write $Y = E(X|\mathcal{G})$.

Note 6.1.0.3. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $(S, \mathcal{S})$ a measurable space, $X \in L^1(\Omega, \mathcal{F}, P)$ and $Y \in L^0_S(\Omega, \mathcal{F})$. We typically write $E(X|Y)$ instead of $E(X|Y^*\mathcal{S})$.

Exercise 6.1.0.4. Existence of Conditional Expectation:
Let $(\Omega, \mathcal{F}, P)$ be a probability space, $\mathcal{G}$ a sub σ-algebra of $\mathcal{F}$ and $X \in L^1(\Omega, \mathcal{F}, P)$. Define Q and $P_{\mathcal{G}}$ as in the previous exercise. Define $Y = dQ/dP_{\mathcal{G}}$. Then Y is a conditional expectation of X given $\mathcal{G}$.

Proof. The Radon-Nikodym theorem implies that Y is $\mathcal{G}$-measurable. Since Q is finite, so is $|Q|$. Since $d|Q| = |Y| \, dP_{\mathcal{G}}$, we have that $Y \in L^1(\Omega, \mathcal{G}, P_{\mathcal{G}})$. An exercise in section 3.3 of [4], implies that for each $G \in \mathcal{G}$

$$\begin{aligned}\int_G Y \, dP &= \int_G Y \, dP_{\mathcal{G}} \\ &= Q(G) \\ &= \int_G X \, dP\end{aligned}$$

□

Definition 6.1.0.5. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $(S, \mathcal{S})$ a measurable space, $X \in L^1(\Omega, \mathcal{F}, P)$ and $Y \in L^0_S(\Omega, \mathcal{F})$. Let $\phi \in L^0(Y(\Omega), \mathcal{S} \cap Y(\Omega))$. Then ϕ is said to be a **conditional expectation function of X given Y** if for each $B \in \mathcal{S} \cap Y(\Omega)$,

$$\int_{Y^{-1}(B)} X \, dP = \int_B \phi \, dP_Y$$

To denote this, we write $\phi(y) = E[X|Y = y]$.

Exercise 6.1.0.6. Existence of Conditional Expectation Function:
Let $(\Omega, \mathcal{F}, P)$ be a probability space, $(S, \mathcal{S})$ a measurable space, $X \in L^1(\Omega, \mathcal{F}, P)$ and $Y \in L^0_S(\Omega, \mathcal{F})$. Suppose that for each $y \in S$, $\{y\} \in \mathcal{S}$. Then there exists $\phi \in L^0(Y(S), \mathcal{S} \cap Y(\Omega))$ such that ϕ is a conditional expectation function of X given Y.
Hint: Doob-Dynkin lemma

Proof. Since $E[X|Y] \in L^0(\Omega, Y^*\mathcal{S})$, the Doob-Dynkin lemma implies that there exists $\phi \in L^0(Y(\Omega), \mathcal{S} \cap Y(\Omega))$ such that $\phi \circ Y = E(X|Y)$. Let $B \in \mathcal{S} \cap Y(\Omega)$. Then

$$\begin{aligned}\int_B \phi \, dP_Y &= \int_{Y^{-1}(B)} \phi \circ Y \, dP \\ &= \int_{Y^{-1}(B)} E(X|Y) \, dP \\ &= \int_{Y^{-1}(B)} X \, dP\end{aligned}$$

□

6.2 Conditional Probability

Definition 6.2.0.1. Let $(A, \mathcal{A})$ be a measurable space, $(B, \mathcal{B}, P_Y)$ a probability space and $Q : B \times \mathcal{A} \to [0, 1]$. Then Q is said to be a **stochastic transition kernel from** $(B, \mathcal{B}, P)$ **to** $(A, \mathcal{A})$ if

1. for each $E \in \mathcal{A}$, $Q(\cdot, E)$ is $\mathcal{B}$-measurable

2. for P-a.e. $b \in B$, $Q(b, \cdot)$ is a probability measure on $(A, \mathcal{A})$

Definition 6.2.0.2. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X, Y \in L^0_n(\Omega, \mathcal{F}, P)$ and $Q : \mathbb{R}^n \times \mathcal{F} \to [0, 1]$. Then Q is said to be a **conditional probability distribution of** X **given** Y if

1. Q is a stochastic transition kernel from $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), P_Y)$ to $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$

2. for each $A, B \in \mathcal{F}$,
$$\int_B Q(y, A) dP_Y(y) = P(X \in A, Y \in B)$$

Note 6.2.0.3. It is helpful to connect this notion of conditional probability with the elementary one by writing $Q(y, A) = P(X \in A|Y = y)$. If $P_Y \ll \mu$, then property (2) in the definition becomes

$$\begin{aligned}P(X \in A, Y \in B) &= \int_B Q(y, A) dP_Y(y) \\ &= \int_B P(X \in A|Y = y) f_Y(y) d\mu(y)\end{aligned}$$

as in a first course on probability.

Exercise 6.2.0.4. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X, Y \in L^0_n$ and $Q : \mathbb{R}^n \times \mathcal{F} \to [0, 1]$. Suppose that for each $A \in \mathcal{F}$, $Q(\cdot, A)$ is $\mathcal{B}(\mathbb{R}^n)$-measurable, for P_Y-a.e. $y \in \mathbb{R}^n$, $P_{X|Y}(y, \cdot)$ is a probability measure on $(\Omega, \mathcal{F})$ and $Q(Y, A) = P(X \in A|Y)$ a.e. Then Q is a conditional probability of X given Y.

Proof. By assumption, for each $A \in \mathcal{F}$, $Q(\cdot, A)$ is $\mathcal{B}(\mathbb{R}^n)$-measurable and for P_Y-a.e. $y \in \mathbb{R}^n$, $Q(y, \cdot)$ is a probability measure on $(\Omega, \mathcal{F})$. Let $A, B \in \mathcal{F}$. Then

$$\begin{aligned}
\int_B Q(y, A) dP_Y(y) &= \int_{Y^{-1}(B)} Q(Y(\omega), A) dP(\omega) \\
&= \int_{Y^{-1}(B)} P(X \in A|Y) dP \\
&= \int_{Y^{-1}(B)} E[1_{X^{-1}(A)}|Y] dP \\
&= \int_{Y^{-1}(B)} 1_{X^{-1}(A)} dP \\
&= \int 1_{X^{-1}(A)} 1_{Y^{-1}(B)} dP \\
&= \int 1_{X^{-1}(A) \cap Y^{-1}(B)} dP \\
&= P(X \in A, Y \in B)
\end{aligned}$$

So Q is a conditional probability distribution of X given Y. □

Definition 6.2.0.5. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X, Y \in L^0_n$ and μ a σ-finite measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Suppose that $P_X, P_Y \ll \mu$. Then $P_{X,Y} \ll \mu^2$. Let $f_X = dP_X/d\mu$, $f_Y = dP_Y/d\mu$ and $f_{X,Y} = dP_{X,Y}/d\mu^2$. Define $f_{X|Y} : \mathbb{R}^n \times \mathbb{R}^n$ by

$$f_{X|Y}(x, y) = \begin{cases} \frac{f_{X,Y}(x,y)}{f_Y(y)}, & y \in \operatorname{supp} Y \\ 0, & y \notin \operatorname{supp} Y \end{cases}$$

Then $f_{X|Y}$ is called the **conditional probability density of X given Y**.

Exercise 6.2.0.6. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X, Y \in L^0_n$ and μ a σ-finite measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Suppose that $P_X, P_Y \ll \mu$. Define $Q : \mathbb{R}^n \times \mathcal{F} \to [0, 1]$ by

$$Q(y, A) = \int_A f_{X|Y}(x, y) d\mu(x)$$

Then Q is a conditional probability distribution of X given Y.

Proof. By the Fubini-Tonelli Theorem, for each $A \in \mathcal{F}$, $Q(\cdot, A)$ is $\mathcal{B}(\mathbb{R}^n)$-measurable and for P_Y-a.e. $y \in \mathbb{R}^n$, $Q(y, \cdot)$ is a probability measure on $(\Omega, \mathcal{F})$. Let $A, B \in \mathcal{F}$. Then

$$\begin{aligned}
\int_B Q(y, A) dP_Y(y) &= \int_B \left[\int_A f_{X|Y}(x, y) d\mu(x) \right] dP_Y(y) \\
&= \int_{B \cap \operatorname{supp} Y} \left[\int_{A \cap \operatorname{supp} Y} \frac{f_{X,Y}(x, y)}{f_Y(y)} d\mu(x) f_Y(y) \right] d\mu(y) \\
&= \int_{B \cap \operatorname{supp} Y} \left[\int_A f_{X,Y}(x, y) d\mu(x) \right] d\mu(y) \\
&= P(X \in A, Y \in B \cap \operatorname{supp} Y) \\
&= P(X \in A, Y \in B)
\end{aligned}$$

□

Theorem 6.2.0.7. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X, Y \in L^1_n(\Omega, \mathcal{F}, P)$. Suppose that $\operatorname{Img} X \in \mathcal{B}(\mathbb{R}^n)$. Then there exists a conditional probability distribution of Y given X.

Chapter 7

Markov Chains

Definition 7.0.0.1. Define Markov kernels, their composition and

Definition 7.0.0.2. Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(X_n)_{n\in\mathbb{N}_0} \in L^0_n$. Then $(X_n)_{n\in\mathbb{N}_0}$ is said to be a **homogeneous Markov chain** if for each $A \in \mathcal{F}$ and $n \in \mathbb{N}$, $P(X_n \in A|X_1, \cdots, X_{n-1}) = P(X_1 \in A|X_0)$ a.e.

Chapter 8

Probabilities Induced by Isometric Group Actions

8.1 Applications to Bayesian Statistics

Exercise 8.1.0.1. Let $(\mathcal{X}, \mathcal{A})$ be a measurable space (Θ, d) a metric space, G a group, $\phi : G \times \Theta \to \Theta$ an isometric group action. Suppose that $\bar{d}$ is a metric on Θ/G. Let

- H_p^Θ be the Hausdorff measure on Θ, $\mu_{\mathcal{X}}$ a measure on $\mathcal{X}$,
- p a denisty on Θ and for each $\theta \in \Theta$, $p(\cdot|\theta)$ a density on $\mathcal{X}$.
- $\theta_0 \in \Theta$ and for $j \in \mathbb{N}$, $X_j \sim p(x|\theta_0)$

Suppose that μ_Θ is G-invariant, p is G-invariant and continuous on Θ and for each $x \in \mathcal{X}$, $p(x|\cdot)$ is G-invariant and continuous on Θ. For $n \in \mathbb{N}$, set $p(\cdot|X^{(n)}) \propto f(X_1, \ldots, X_n|\cdot)p(\cdot)$. Define the posterior measure $P_{\Theta|X^{(n)}} : \mathcal{B}(\Theta) \to [0, 1]$ by

$$dP_{\Theta|X^{(n)}}(\theta) = p(\theta|X^{(n)})\, dH_p^\Theta(\theta)$$

Then there exists a density $\bar{p}(\cdot|X^{(n)})$ on Θ/G such that

$$d\bar{P}_{\Theta|X^{(n)}}(\theta) = \bar{p}(\theta|X^{(n)})\, d\bar{H}^\Theta(\theta)$$

Proof. Clear from previous work. □

Exercise 8.1.0.2. Let $(\mathcal{X}, \mathcal{A})$ be a measurable space (Θ, d) a metric space, G a group, $\phi : G \times \Theta \to \Theta$ an isometric group action. Suppose that $\bar{d}$ is a metric on Θ/G. Let

- H_p^Θ be the Hausdorff measure on Θ, $\mu_{\mathcal{X}}$ a measure on $\mathcal{X}$,
- p a denisty on Θ and for each $\theta \in \Theta$, $p(\cdot|\theta)$ a density on $\mathcal{X}$.
- $\theta_0 \in \Theta$ and for $j \in \mathbb{N}$, $X_j \sim p(x|\theta_0)$

Suppose p is G-invariant and continuous on Θ and for each $x \in \mathcal{X}$, $p(x|\cdot)$ is G-invariant and continuous on Θ. For $n \in \mathbb{N}$, set $p(\cdot|X^{(n)}) \propto f(X_1, \ldots, X_n|\cdot)p(\cdot)$. Define the posterior measure $P_{\Theta|X^{(n)}} : \mathcal{B}(\Theta) \to [0, 1]$ by

$$dP_{\Theta|X^{(n)}}(\theta) = p(\theta|X^{(n)})\, dH_p^\Theta(\theta)$$

Suppose that $(P_{\Theta|X^{(n)}})_{n\in\mathbb{N}}$ concentrates on $\bar{\theta}_0 \subset \Theta$ a.s. or in probability. Then $(\bar{P}_{\Theta|X^{(n)}})_{n\in\mathbb{N}}$ concentrates a.s. or in probability on $\{\bar{\theta}_0\} \subset \Theta/G$ (i.e. is consistent a.s. or in probability)

Proof. Let $V \in \mathcal{N}_{\bar{\theta}_0}$. Then $\pi^{-1}(V) \in \mathcal{N}_{\bar{\theta}_0}$. By definition,

$$\begin{aligned}\bar{P}_{\Theta|X^{(n)}}(V^c) &= P_{\Theta|X^{(n)}}(\pi^{-1}(V^c)) \\ &= P_{\Theta|X^{(n)}}(\pi^{-1}(V)^c) \\ &\xrightarrow{\text{a.s.}/P} 0\end{aligned}$$

□

Note 8.1.0.3. Some examples of G-invariant priors would be the uniform distribution, or $N_n(0, \sigma^2 I)$ on $\mathbb{R}^n$ when acted on by $O(n)$. An example of a G-invariant likelihood would be $f(A|Z) \sim \text{Ber}(ZZ^T)$ as in a latent position random graph model where $Z \in \mathbb{R}^{n \times d}$ is the parameter is invariant under right multiplication by $U \in O_d$.

Chapter 9

Projective Systems

Chapter 10

Stochastic Integration

Exercise 10.0.0.1. Let $(\Omega, \mathcal{F}, P)$ be a probability space, X a set $\mathcal{A}_0$ an algebra, $\mu_0 : \mathcal{A}_0 \to \mathbb{C}$ and $B : \mathcal{A}_0 \to L^2(\Omega, \mathcal{F}, P)$. Suppose that

1. $B(\varnothing) = 0$
2. for each $E, F \in \mathcal{A}_0$, if $E \cap F = \varnothing$, then $B(E \cup F) = B(E) + B(F)$
3. $E(B(E)B(F)^*) = \mu_0(E \cap F)$

Then

1. for each $E \in \mathcal{A}_0$, $\mu_0(E) = E(|B(E)|^2)$.
2. for each $E \in \mathcal{A}_0$, $0 \leq \mu_0(E) < \infty$
3. for each $E, F \in \mathcal{A}_0$, if $E \cap F = \varnothing$, then $\mu_0(E \cup F) = \mu_0(E) + \mu_0(F)$

Proof.

1. Clear
2. Clear
3. Let $E, F \in \mathcal{A}_0$. Suppose that $E \cap F = \varnothing$. Then

$$\begin{aligned} E(B(E)B(F)^*) &= \mu_0(E \cap F) \\ &= \mu_0(\varnothing) \\ &= E(|B(\varnothing)|^2) \\ &= E(0) \\ &= 0 \end{aligned}$$

This implies that

$$\begin{aligned} \mu_0(E \cup F) &= E(|B(E \cup F)|^2) \\ &= E(|B(E) + B(F)|^2) \\ &= E(|B(E)|^2) + E(|B(F)|^2) + 2\mathrm{Re}\, E(B(E)B(F)^*) \\ &= \mu_0(E) + \mu_0(F) + 0 \\ &= \mu_0(E) + \mu_0(F) \end{aligned}$$

□

Definition 10.0.0.2. Let $(\Omega, \mathcal{F}, P)$ be a probability space, X a set $\mathcal{A}_0$ an algebra, $\mu_0 : \mathcal{A}_0 \to [0, \infty)$ a premeasure and $B : \mathcal{A}_0 \to L^2(\Omega, \mathcal{F}, P)$. Suppose that

1. $B(\varnothing) = 0$
2. for each $E, F \in \mathcal{A}_0$, if $E \cap F = \varnothing$, then $B(E \cup F) = B(E) + B(F)$
3. $E(B(E)B(F)^*) = \mu_0(E \cap F)$

Then B is said to be a **stochastic premeasure with sturcture** μ_0

Chapter 11

TODO

Incorporate vector measures to have conditional expectations of Banach space valued functions given a σ-algebra

Appendix A

Summation

Definition A.0.0.1. Let $f : X \to [0, \infty)$, Then we define

$$\sum_{x \in X} f(x) := \sup_{\substack{F \subset X \\ F \text{ finite}}} \sum_{x \in F} f(x)$$

This definition coincides with the usual notion of summation when X is countable. For $f : X \to \mathbb{C}$, we can write $f = g + ih$ where $g, h : X \to \mathbb{R}$. If

$$\sum_{x \in X} |f(x)| < \infty,$$

then the same is true for g^+, g^-, h^+, h^-. In this case, we may define

$$\sum_{x \in X} f(x)$$

in the obvious way.

The following note justifies the notation $\sum_{x \in X} f(x)$ where $f : X \to \mathbb{C}$.

Note A.0.0.2. Let $f : X \to \mathbb{C}$ and $\alpha : X \to X$ a bijection. If $\sum_{x \in X} |f(x)| < \infty$, then $\sum_{x \in X} f(\alpha(x)) = \sum_{x \in X} f(x)$.

Appendix B

Asymptotic Notation

Definition B.0.0.1. Let X be a topological space, Y, Z be normed vector spaces, $f : X \to Y$, $g : X \to Z$ and $x_0 \in X \cup \{\infty\}$. Then we write

$$f = o(g) \qquad \text{as } x \to x_0$$

if for each $\epsilon > 0$, there exists $U \in \mathcal{N}(x_0)$ such that for each $x \in U$,

$$\|f(x)\| \leq \epsilon \|g(x)\|$$

Exercise B.0.0.2. Let X be a topological space, Y, Z be normed vector spaces, $f : X \to Y$, $g : X \to Z$ and $x_0 \in X \cup \{\infty\}$. If there exists $U \in \mathcal{N}(x_0)$ such that for each $x \in U \setminus \{x_0\}$, $g(x) > 0$, then

$$f = o(g) \text{ as } x \to x_0 \quad \text{iff} \quad \lim_{x \to x_0} \frac{\|f(x)\|}{\|g(x)\|} = 0$$

Exercise B.0.0.3. Let X and Y a be normed vector spaces, $A \subset X$ open and $f : A \to Y$. Suppose that $0 \in A$. If $f(h) = o(\|h\|)$ as $h \to 0$, then for each $h \in X$, $f(th) = o(|t|)$ as $t \to 0$.

Proof. Suppose that $f(h) = o(\|h\|)$ as $h \to 0$. Let $h \in X$ and $\epsilon > 0$. Choose $\delta' > 0$ such that for each $h' \in B(0, \delta')$, $h' \in A$ and

$$\|f(h')\| \leq \frac{\epsilon}{\|h\| + 1} \|h'\|$$

Choose $\delta > 0$ such that for each $t \in B(0, \delta)$, $th \in B(0, \delta')$. Let $t \in B(0, \delta)$. Then

$$\begin{aligned} \|f(th)\| &\leq \frac{\epsilon}{\|h\| + 1} |t| \|h\| \\ &< \epsilon |t| \end{aligned}$$

So $f(th) = o(|t|)$ as $t \to 0$. $\square$

Definition B.0.0.4. Let X be a topological space, Y, Z be normed vector spaces, $f : X \to Y$, $g : X \to Z$ and $x_0 \in X \cup \{\infty\}$. Then we write

$$f = O(g) \qquad \text{as } x \to x_0$$

if there exists $U \in \mathcal{N}(x_0)$ and $M \geq 0$ such that for each $x \in U$,

$$\|f(x)\| \leq M \|g(x)\|$$

Appendix C

Categories

move to notation?

Definition C.0.0.1. We define the category of topological measure spaces, denoted $\mathbf{TopMsr}_+$, by

- $\mathrm{Obj}(\mathbf{TopMsr}_+) := \{(X, \mu) : X \in \mathrm{Obj}(\mathbf{Top}) \text{ and } \mu \in M(X)\}$
- $\mathrm{Hom}_{\mathbf{TopMsr}_+}((X, \mu), (Y, \nu)) := \mathrm{Hom}_{\mathbf{Top}}(X, Y) \cap \mathrm{Hom}_{\mathbf{Msr}_+}((X, \mathcal{B}(X), \mu), (Y, \mathcal{B}(Y), \nu))$

Appendix D

Rings

Appendix E

Introduction

Definition E.0.0.1. Let R be a ring and $I, J \subset R$ ideals. We define

- $I + J \subset R$ by $I + J := \{i + j : i \in I \text{ and } j \in J\}$,
- $IJ \subset R$ by $IJ := \{i_1 j_1 + i_2 j_2 : i_1, i_2 \in I \text{ and } j_1, j_2 \in J\}$.

Exercise E.0.0.2. Let R be a ring and $I, J \subset R$ ideals. Then $I + J$ and IJ are ideals.

Proof. FINISH!!! □

Appendix F

Quotient Rings

Exercise F.0.0.1. Let R be a commutative unital ring and $I \subset R$ and ideal. Then I is maximal iff R/I is a field.

Proof. A previous exercise implies that R/I is a commutative unital ring.

- $(\implies)$:
  Suppose that I is maximal. Let $k + I \in R/I$. Suppose that $k + I \neq 0 + I$. Then $k \notin I$. Define $J \subset R$ by $J := (k) + I$. A previous exercise implies that J is an ideal. By construction $I \subset J$ and $k \in J$. Hence $I \neq J$. Since I is maximal, $J = R$. Since $1 \in J$, there exists $r \in R$ and $i \in I$ such that $1 = rk + i$. Hence

  $$\begin{aligned} 1 + I &= rk + i + I \\ &= rk + I \\ &= (r + I)(k + I). \end{aligned}$$

  Thus $(k+I)$ is invertible. Since $k+I \in R/I$ such that $k+I \neq 0+I$ is arbitrary, we have that for each $k+I \in R/I$, $k+I \neq 0+I$ implies that $k + I$ is invertible. Thus R/I is a field.

- $(\impliedby)$:
  Suppose that R/I is a field. Let $J \subset R$. Suppose that J is an ideal, $I \subset J$ and $I \neq J$. Then there exists $j \in J$ such that $j \notin I$. Thus $j + I \neq 0 + I$. Since R/I is a field, $j + I$ is invertible. Hence there exists $k \in R$ such that $jk + I = 1 + I$. Hence

  $$\begin{aligned} 1 - jk &\in I \\ &\subset J. \end{aligned}$$

  Since J is an ideal, $jk \in J$ and therefore

  $$\begin{aligned} 1 &= (1 - jk) + jk \\ &\in J. \end{aligned}$$

  Since $1 \in J$, $J = R$. Since $J \subset R$ with J an ideal and $I \subset J$ is arbitrary, we have that for each $J \subset R$, J an ideal and $I \subset J$ implies that $J = I$ or $J = X$. Hence I is maximal.

$\square$

Appendix G

Vector Spaces

it might be better to cover some category theory and write everything in terms of $\mathrm{Hom}_{\mathbf{Vect}_{\mathbb{K}}}$ and $\mathrm{Obj}(\mathbf{Vect}_{\mathbb{K}})$

Appendix H

Introduction

Definition H.0.0.1. Let X be a set, $\mathbb{K}$ a field, $+ : X \times X \to X$ and $\cdot : \mathbb{K} \times X \to X$. Then $(X, +, \cdot)$ is said to be a **$\mathbb{K}$-vector space** if

1. $(X, +)$ is an abelian group
2.

Definition H.0.0.2. Let $(X, +, \cdot) \in \mathrm{Obj}(\mathbf{Vect}_{\mathbb{C}})$. We define the **conjugate of** $(X, +, \cdot)$, denoted $(\bar{H}, \bar{+}, \bar{\cdot})$, by

- $\bar{X} := X$,
- $x \bar{+} y := x + y$,
- $\lambda \bar{\cdot} x := \lambda^* \cdot x$.

Exercise H.0.0.3. Let $(X, +, \cdot) \in \mathrm{Obj}(\mathbf{Vect}_{\mathbb{C}})$. Then $(\bar{H}, \bar{+}, \bar{\cdot}) \in \mathrm{Obj}(\mathbf{Vect}_{\mathbb{C}})$.

Proof. FINISH!!! □

Definition H.0.0.4. Let $(X, +_X, \cdot_X)$ and $(E, +_E, \cdot_E)$ be vector spaces. Suppose that $E \subset X$. Then $(E, +_E, \cdot_E)$ is said to be a subspace of X if

1. $+_E = +_X|_{E \times E}$
2. $\cdot_E = \cdot_X|_{\mathbb{K} \times E}$

Exercise H.0.0.5. Let $(X, +_X, \cdot_X)$ and $(E, +_E, \cdot_E)$ be vector spaces. Suppose that $E \subset X$.

Exercise H.0.0.6. Let $(X, +, \cdot)$ be a vector space and $E \subset X$. Then E is a subspace of X

Definition H.0.0.7. Let X be a vector space and $(E_j)_{j \in J}$ a collection of subspaces of X. Then $\bigcap\limits_{j \in J} E_j$ is a subspace of X.

Proof. Set $E := \bigcap\limits_{j \in J} E_j$. Let $x, y \in E$ and $\lambda \in \mathbb{K}$. Then for each $j \in J$, $x, y \in E_j$. Since for each $j \in J$, E_j is a subspace of X, we have that for each $j \in J$, $x + \lambda y \in E_j$. Thus $x + \lambda y \in E$. Since $x, y \in E$ and $\lambda \in \mathbb{K}$ are arbitrary, (cite exercise here) we have that E is a subspace of X. □

Definition H.0.0.8. Let X, Y be vector spaces and $T : X \to Y$. Then T is said to be **linear** if for each $x_1, x_2 \in X$ and $\lambda \in \Lambda$,

1. $T(x_1 + x_2) = T(x_1) + T(x_2)$,
2. $T(\lambda x_1) = \lambda T(x_1)$.

We define $\mathcal{L}(X; Y) := \{T : X \to Y : \ T \text{ is linear}\}$.

Exercise H.0.0.9. Let X, Y be vector spaces and $T : X \to Y$. Then T is linear iff for each $x_1, x_2 \in X$ and $\lambda \in \Lambda$,

$$T(x_1 + \lambda x_2) = T(x_1) + \lambda T(x_2)$$

Proof. Clear. (add details) □

Exercise H.0.0.10. $\mathcal{L}(X; Y)$ is a vector space FINISH!!!

Proof. content... □

Definition H.0.0.11. define addition/scalar multiplication of linear maps

Exercise H.0.0.12. Let X, Y be vector spaces over $\mathbb{K}$. Then $\mathcal{L}(X; Y)$ is a $\mathbb{K}$-vector space.

Proof. Clear □

Exercise H.0.0.13. Let X, Y, Z be vector spaces over $\mathbb{K}$, $T \in \mathcal{L}(X.Y)$ and $S \in \mathcal{L}(Y, Z)$. Then $S \circ T \in \mathcal{L}(X, Z)$.

Proof. FINISH!!! □

Appendix I

Bases

Definition I.0.0.1. Let X be a vector space and $(e_\alpha)_{\alpha \in A} \subset X$. Then $(e_\alpha)_{\alpha \in A}$ is said to be

- **linearly independent** if for each $(\alpha_j)_{j=1}^n \subset A$, $(\lambda_j)_{j=1}^n \subset \mathbb{K}$, $\sum\limits_{j=1}^n \lambda_j e_{\alpha_j} = 0$ implies that for each $j \in [n]$, $\lambda_j = 0$.
- a **Hamel basis for** X if $(e_\alpha)_{\alpha \in A}$ is linearly independent and $\operatorname{span}(e_\alpha)_{\alpha \in A} = X$.

Exercise I.0.0.2. every vector space has a Hamel basis

Proof. □

Exercise I.0.0.3.

Exercise I.0.0.4. Let X be a $\mathbb{K}$-vector space and $x \in X$. Then $x = 0$ iff for each $\phi \in X^*$, $\phi(x) = 0$.

Proof.

- $(\implies)$:
  Suppose that $x = 0$. Linearity implies that for each $\phi \in X^*$ $\phi(x) = 0$.
- $(\impliedby)$:
  Conversely, suppose that $x \neq 0$. Define $\epsilon_x : \operatorname{span}(x) \to \mathbb{K}$ by $\epsilon_x(\lambda x) := \lambda$. Let $u, v \in \operatorname{span}(x)$. Then there exists $\lambda_u, \lambda_v \in \mathbb{K}$ such that $u = \lambda_u x$ and $v = \lambda_v x$. Suppose that $u = v$. Then

  $$\begin{aligned}(\lambda_u - \lambda_v)x &= \lambda_u x - \lambda_v x \\ &= u - v \\ &= 0\end{aligned}$$

  Since $x \neq 0$, we have that $\lambda_u - \lambda_v = 0$ and therefore $\lambda_u = \lambda_v$. Hence

  $$\begin{aligned}\lambda_u &= \epsilon_x(u) \\ &= \epsilon_x(v) \\ &= \lambda_v.\end{aligned}$$

  Thus ϵ_x is well defined.

□

Appendix J

Duality

Definition J.0.0.1. Let X be a vector space over $\mathbb{K}$. We define the **algebraic dual space of** X, denoted X', by $X' := \mathcal{L}(X; \mathbb{K})$.

Exercise J.0.0.2. Let X be a vector space. Then X' is a vector space.

Proof. Clear. $\square$

Definition J.0.0.3. Let X, Y be vector spaces over $\mathbb{K}$. We define the algebraic adjoint of $T \in \mathcal{L}(X; Y)$, denoted $T' \in \mathcal{L}(Y', X')$, by $T'(\phi) := \phi \circ T$.

Appendix K

Multilinear Maps

Definition K.0.0.1. Let $X_1, \cdots, X_n, Y$ be vector spaces and $T : \prod_{j=1}^{n} X_j \to \mathbb{K}$. Then T is said to be **multilinear** if for each $j_0 \in [n]$ and $(x_j)_{j=1}^n \in \prod_{j=1}^{n} X_j$, $T(x_1, \ldots, x_{j_0-1}, \cdot, x_{j_0+1})$ is linear.

$$L^n(X_1, \ldots, X_n; Y) = \left\{ T : \prod_{j=1}^{n} X_j \to Y : T \text{ is multilinear} \right\}$$

If $X_1 = \cdots = X_n = X$, we write $L^n(X; Y)$ in place of $L^n(X, \ldots, X; Y)$.

Definition K.0.0.2. define addition and scalar mult of multilinear maps

Exercise K.0.0.3. Let $X_1, \cdots, X_n, Y$ be vector spaces. Then $L^n(X_1, \ldots, X_n; Y)$ is a $\mathbb{K}$-vector space.

Proof. content... □

Exercise K.0.0.4. Let $X_1, \cdots, X_n, Y, Z$ be $\mathbb{K}$-vector spaces, $\alpha \in L^n(X_1, \ldots, X_n; Y)$ and $\phi \in L^1(Y; Z)$. Then $\phi \circ \alpha \in L^n(X_1, \ldots, X_n; Z)$.

Proof. Let $(x_j)_{j=1}^n \in \prod_{j=1}^{n} X_j$ and $j_0 \in [n]$. Define $f : X_{j_0} \to Y$ by

$$f(a) := \alpha(x_1, \ldots, x_{j_0-1}, a, x_{j_0+1}, \ldots, x_n)$$

Since $\alpha \in L^n(X_1, \ldots, X_n; Y)$, f is linear. Since ϕ is linear, and $\phi \circ f$ is linear. Since $(x_j)_{j=1}^n \in \prod_{j=1}^{n} X_j$ and $j_0 \in [n]$ are arbitrary, we have that $\phi \circ \alpha \in L^n(X_1, \ldots, X_n; Y)$. □

Appendix L

Tensor Products

Definition L.0.0.1. Let X, Y and T be vector spaces over $\mathbb{K}$ and $\alpha \in L^2(X, Y; T)$. Then (T, α) is said to be a **tensor product of X and Y** if for each vector space Z and $\beta \in L^2(X, Y; Z)$, there exists a unique $\phi \in L^1(T; Z)$ such that $\phi \circ \alpha = \beta$, i.e. the following diagram commutes:

$$\begin{tikzcd} X \times Y \arrow[r, "\alpha"] \arrow[dr, "\beta"'] & T \arrow[d, dashed, "\phi"] \\ & Z \end{tikzcd}$$

Exercise L.0.0.2. Let X, Y, S, T be vector spaces, $\alpha \in L^2(X, Y; S)$ and $\beta \in L^2(X, Y; T)$. Suppose that (S, α) and (T, β) are tensor products of X and Y. Then S and T are isomorphic.

Proof. Since (T, β) is a tensor product of X and Y, $\beta \in L^2(X, Y; T)$ there exists a unique $f \in L^1(T; T)$ such that $f \circ \beta = \beta$, i.e. the following diagram commutes:

$$\begin{tikzcd} & T \arrow[dd, dashed, "f"] \\ X \times Y \arrow[ur, "\beta"] \arrow[dr, "\beta"'] & \\ & T \end{tikzcd}$$

Since $\mathrm{id}_T \in L^1(T; T)$ and $\mathrm{id}_T \circ \beta = \beta$, we have that $f = \mathrm{id}_T$. Since (S, α) is a tensor product of X and Y, there exists a unique $\phi : S \to T$ such that $\phi \circ \alpha = \beta$, i.e. the following diagram commutes:

$$\begin{tikzcd} X \times Y \arrow[r, "\alpha"] \arrow[dr, "\beta"'] & S \arrow[d, dashed, "\phi"] \\ & T \end{tikzcd}$$

Similarly, since (T, β) is a tensor product of X and Y, there exists a unique $\psi : T \to S$ such that $\psi \circ \beta = \alpha$, i.e. the following diagram commutes:

$$\begin{tikzcd} X \times Y \arrow[r, "\beta"] \arrow[dr, "\alpha"'] & T \arrow[d, dashed, "\psi"] \\ & S \end{tikzcd}$$

Therefore

$$\begin{aligned} (\phi \circ \psi) \circ \beta &= \phi \circ (\psi \circ \beta) \\ &= \phi \circ \alpha \\ &= \beta, \end{aligned}$$

i.e. the following diagram commutes:

$$\begin{tikzcd} & T \arrow[d, dashed, "\psi"] \\ X \times Y \arrow[ur, "\beta"] \arrow[r, "\alpha"] \arrow[dr, "\beta"'] & S \arrow[d, dashed, "\phi"] \\ & T \end{tikzcd} \implies \begin{tikzcd} & T \arrow[dd, dashed, "\phi\circ\psi"] \\ X \times Y \arrow[ur, "\beta"] \arrow[dr, "\beta"'] & \\ & T \end{tikzcd}$$

By uniqueness of $f \in L^1(T;T)$, we have that

$$\begin{aligned} \mathrm{id}_T &= f \\ &= \phi \circ \psi \end{aligned}$$

A similar argument implies that $\psi \circ \phi = \mathrm{id}_S$. Hence ϕ and ψ are isomorphisms with $\phi^{-1} = \psi$. Hence S and T are isomorphic. □

Definition L.0.0.3. Let X, Y be vector spaces, $x \in X$ and $y \in Y$. We define $x \otimes y : X^* \times Y^* \to \mathbb{K}$ by $x \otimes y(\phi, \psi) := \phi(x)\psi(y)$.

Exercise L.0.0.4. Let X, Y be vector spaces, $x \in X$ and $y \in Y$. Then $x \otimes y \in L^2(X^*, Y^*; \mathbb{K})$.

Proof. Let $\phi_1, \phi_2 \in X^*$, $\psi \in Y^*$ and $\lambda \in \mathbb{K}$. Then

$$\begin{aligned} x \otimes y(\phi_1 + \lambda\phi_2, \psi) &= [\phi_1 + \lambda\phi_2(x)]\psi(y) \\ &= \phi_1(x)\psi(y) + \lambda\phi_2(x)\psi(y) \\ &= x \otimes y(\phi_1, \psi) + \lambda x \otimes y(\phi_2, \psi) \end{aligned}$$

Since $\phi_1, \phi_2 \in X^*$, $\psi \in Y^*$ and $\lambda \in \mathbb{K}$ are arbitrary, we have that for each $\psi \in Y^*$, $x \otimes y(\cdot, \psi)$ is linear. Similarly for each $\phi \in X^*$, $x \otimes y(\phi, \cdot)$ is linear. Hence $x \otimes y$ is bilinear and $x \otimes y \in L^2(X^*, Y^*; \mathbb{K})$. □

Definition L.0.0.5. Let X, Y be vector spaces. We define

- the **tensor product of X and Y**, denoted $X \otimes Y \subset L^2(X^*, Y^*; \mathbb{K})$, by

$$X \otimes Y := \mathrm{span}(x \otimes y : x \in X \text{ and } y \in Y),$$

- the **tensor map**, denoted $\otimes : X \times Y \to X \otimes Y$, by $\otimes(x, y) := x \otimes y$.

Exercise L.0.0.6. Let X, Y be vector spaces, $(x_j)_{j=1}^n \subset X$ and $(y_j)_{j=1}^n \subset Y$. The following are equivalent:

1. $\sum_{j=1}^n x_j \otimes y_j = 0$
2. for each $\phi \in X^*$ and $\psi \in Y^*$, $\sum_{j=1}^n \phi(x_j)\psi(y_j) = 0$
3. for each $\phi \in X^*$, $\sum_{j=1}^n \phi(x_j)y_j = 0$
4. for each $\psi \in Y^*$, $\sum_{j=1}^n \psi(y_j)x_j = 0$

Proof.

1. $(1) \implies (2)$:
   Suppose that $\sum_{j=1}^n x_j \otimes y_j = 0$. Let $\phi \in X^*$ and $\psi \in Y^*$. Then

$$\begin{aligned} \sum_{j=1}^n \phi(x_j)\psi(y_j) &= \phi\left(\sum_{j=1}^n \psi(y_j)x_j\right) \\ &= \end{aligned}$$

2.

3.

□

Exercise L.0.0.7. Let X, Y be vector spaces. Then $(X \otimes Y, \otimes)$ is a tensor product of X and Y.

Proof. Let Z be a vector space and $\alpha \in L^2(X, Y; Z)$. Define $\phi : X \otimes Y \to Z$ by $\phi\left(\sum_{j=1}^{n} \lambda_j x_j \otimes y_j\right) := \sum_{j=1}^{n} \lambda_j \alpha(x_j, y_j)$.

- **(well defined):**
  Let $u \in X \otimes Y$. Then there exist $(\lambda_j)_{j=1}^n \subset \mathbb{K}$, $(x_j)_{j=1}^n \subset X$, $(y_j)_{j=1}^n \subset Y$ such that $u = \sum_{j=1}^{n} \lambda_j x_j \otimes y_j$. Suppose that $u = 0$. Let $\phi \in Z^*$. Then $\phi \circ \alpha \in L^2(X, Y; Z)$.

□

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